

Unifying Diversity and Fairness in Re-ranking via Economic Growth Theory

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Abstract

Re-ranking is a critical technique for incorporating objectives beyond accuracy, such as fairness and diversity, into recommender systems. However, these objectives often conflict with each other, making their joint optimization a significant challenge. In this paper, we analyze this conflict through the lens of economic redistribution principles: diversity aligns with the Sufficentarian principle, which ensures that each item receives sufficient exposure above a utility threshold, while fairness aligns with the Rawlsian principle, which prioritizes improving the welfare of the least advantaged items. Inspired by this analogy, we design a unified objective function that improves diversity via the Sufficentarian principle, and enhances fairness via the Rawlsian one. However, directly optimizing this objective is NP-hard and requires global information, which poses significant computational challenges. To overcome these challenges, we propose a novel re-ranking algorithm named DivFair, which efficiently optimizes the objective function in online settings. DivFair transforms the original problem into its dual problem, employing a mirror-descent approach and a linear approximation technique that transforms the complex per-round optimization into an efficient sorting task. Theoretical analysis shows that our algorithm guarantees sub-linear regret and stable convergence. Extensive experiments on three large-scale public datasets demonstrate that DivFair outperforms state-of-the-art baselines in balancing accuracy, fairness, and diversity, while maintaining high computational efficiency suitable for online recommendation.

CCS Concepts

• Information systems → Information retrieval.

Keywords

Fairness, Diversity, Re-ranking, Economic Growth

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1 Introduction

Re-ranking techniques are commonly used to ensure a trustworthy recommender system (RS) that incorporates multiple beyond-accuracy objectives, such as fairness [17, 32] and diversity [29, 49], both of which are central to building a responsible web that ensures equitable access to information. Specifically, fairness aims to equalize utility across items [46], whereas diversity seeks to ensure sufficient coverage of distinct items [25]. Although both fairness and diversity benefit RS, prior studies [61] show that jointly optimizing them often leads to inconsistent outcomes. This inconsistency can undermine the effectiveness of both objectives when optimized together.

In this paper, we argue that the relationships between fairness and diversity can be interpreted through the lens of economic growth theory [2, 30]. From an economic perspective, the ranking process [56] in RS mirrors the problem of distribution [16], where resources (e.g., exposure or utility) are allocated based on initial preferences or predicted scores [56]. In contrast, re-ranking [46] resembles the process of redistribution [5], which adjusts the initial allocation to meet certain social or policy goals such as equity or inclusion [47]. This analogy provides a useful framework for understanding how different re-ranking goals align with economic theories of justice [2]. Diversity ensures broad item coverage [25], which aligns with the Sufficentarian principle [24] that maximizes the proportion of the population exceeding a minimum threshold (i.e., the poverty line [40]). Fairness, by contrast, refers to reducing disparities by improving the welfare of the least advantaged [46], which aligns with the Rawlsian principle [42] that prioritizes those worst off in the distribution [45].

We illustrate the above analogy in Figure 1. In the blue area of Figure 1(a), plot (1) shows the regional per-capita GDP imbalance in China in 2000 under a distribution-oriented scheme¹. This

¹Data source: National Bureau of Statistics of China, available at <https://data.stats.gov.cn/english/>

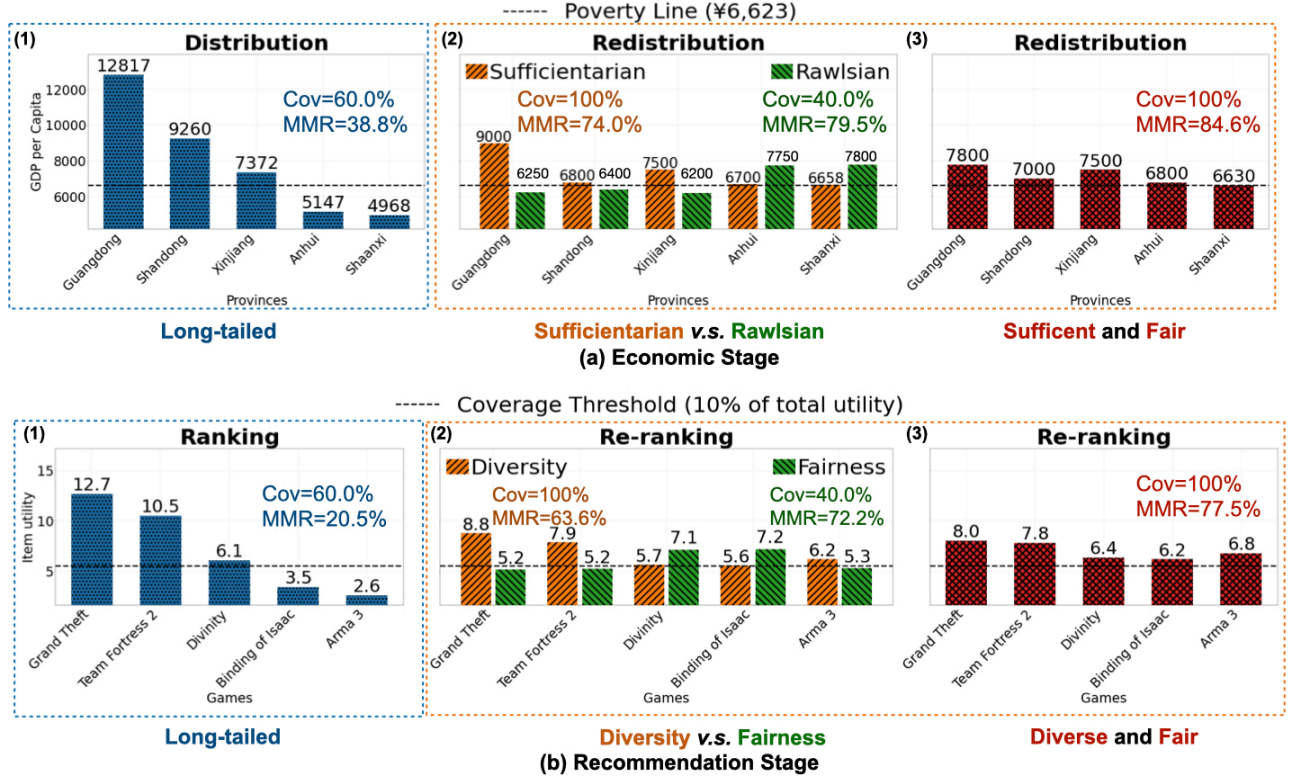


Figure 1: Parallel between (a) Economic Stage and (b) Recommendation Stage. The blue area highlights the similarity between distribution in economics and ranking in RS. The orange area compares redistribution principles (Sufficientarian v.s. Rawlsian) with re-ranking objectives (Diversity v.s. Fairness). Cov and MMR are diversity and fairness metrics, respectively.

long-tailed distribution resembles the ranking outcomes in RS in Figure 1(b), as measured by Cov and MMR, where Cov (coverage) denotes the proportion of distinct items recommended and MMR (min-max ratio) is the ratio of the least to the most exposed item. In the orange area of Figure 1(a), plot (2) visualizes the redistribution process under two policies: a Sufficientarian-oriented policy that lifts as many regions as possible above the poverty line, and a Rawlsian-oriented policy that minimizes regional disparities. These correspond to diversity-oriented and fairness-oriented re-ranking strategies in plot (2) in Figure 1(b). Plot (3) in both (a) and (b) illustrates an ideal outcome where both diversity and fairness are jointly optimized, achieving full coverage and reduced disparity. This analogy enables us to leverage economic theories to jointly optimize fairness and diversity in re-ranking.

Motivated by the iso-elastic utility function in economics [35] and the Bass diffusion model [6], we design an objective function for re-ranking that effectively balances fairness and accuracy. The objective function is piecewise, consisting of a diversity component and a fairness component. The diversity component uses a Bass-derived function that grows exponentially, encouraging utility to reach a coverage threshold and reflecting the Sufficientarian principle. The fairness component incorporates an iso-elastic function whose parameters can recover the Rawlsian principle. However, optimizing this objective poses a challenge: it leads to a non-linear, large-scale integer programming problem [8], which is NP-hard

and computationally expensive, requiring offline optimization.

To reduce complexity and effectively adapt to online settings, we propose a novel re-ranking method, DivFair, which performs efficient online optimization in the dual space through linear approximation. First, we derive its dual form [11] to transform the primal problem into a convex one, and design a mirror-descent algorithm [37] to optimize it. Second, to improve computational efficiency, we propose a linear approximation technique that transforms per-round optimization into a top- K sorting task. Benefiting from the strong concavity and smoothness of the objective function, our theoretical analysis shows that DivFair achieves a sub-linear regret bound.

We summarize the major contributions of this paper as follows:

- (i) We analyze the distinction between fairness and diversity in re-ranking from an economic perspective, and design an objective function that jointly optimizes accuracy, fairness, and diversity.
- (ii) We propose a novel re-ranking model, DivFair, which performs efficient online optimization in the dual space via linear approximation. Theoretical analysis shows that DivFair achieves a sub-linear regret bound and guarantees stable convergence.
- (iii) Extensive experiments on three public datasets demonstrate that our model outperforms state-of-the-art baselines in both re-ranking effectiveness and computational efficiency.

2 Related Work

Fairness in RS. Fairness has become a key focus in re-ranking due to growing concerns about algorithmic bias and its societal consequences [17, 33]. Depending on the stakeholder, existing approaches typically target either user fairness [1, 31] or item fairness [43, 54], with our work focusing on the latter. In multi-stakeholder settings [1], re-ranking methods are particularly appealing for their flexibility and ease of deployment. Re-ranking algorithms can be broadly categorized into (i) regularized methods that balance accuracy and fairness via tunable trade-off parameters [36, 52], and (ii) constraint-based methods that impose fairness thresholds [38, 51]. However, most of these approaches optimize a single objective and incur high computational overhead, limiting scalability in practice.

Diversity in RS. Diversity is another critical concern in RS, motivated by the need to mitigate redundancy and prevent the over-concentration of popular items, potentially reducing user satisfaction and marginalizing long-tail content [3, 62]. Diversity is often studied at two levels: (i) individual diversity, which measures dissimilarity among items in a single user's list, and (ii) aggregate diversity, which evaluates the overall item spread across users [4, 23, 62]. While individual diversity has been widely explored using re-ranking strategies such as Maximum Marginal Relevance (MMR) [12, 62] and Determinantal Point Processes (DPP) [14], aggregate diversity—often measured using coverage [3]—is equally important. Coverage measures the proportion of distinct items recommended across the system. Re-ranking techniques that target aggregate diversity typically apply greedy selection to adjust top-N recommendation lists [4]. In this paper, we focus specifically on improving coverage-based aggregate diversity during the re-ranking phase.

Joint optimization of fairness and diversity. While fairness and diversity have traditionally been addressed separately in RS, recent research has recognized the importance of their joint optimization. InFoDance [60] introduces a data-centric approach using an influence function to enhance accuracy, group fairness, and individual diversity. Similarly, LLM4Rerank [48] introduces a framework based on large language models that treats each optimization goal as a node in a reasoning graph, enabling personalized trade-offs among accuracy, group fairness, and individual diversity. Zhao *et al.* [61] survey the conceptual overlap between fairness and diversity, reinterpreting fairness techniques through the lens of diversity at both user and item levels. In parallel, a recent survey [22] highlights the potential of generative AI (*e.g.*, GANs, VAEs, LLMs) to address multi-objective recommendation. Our work extends this line of inquiry by proposing a unified re-ranking framework that balances item fairness and aggregate diversity in a scalable and controllable manner.

3 Problem Formulation

In this section, we introduce the notations, define item utility and the ranking task, and then formalize diverse re-ranking and fair re-ranking in RS.

RS notations. We denote the user set by $\mathcal{U}$ and the item set by $\mathcal{I}$. For each user $u \in \mathcal{U}$, the recommender system produces a ranked list $L_K(u) \in \mathcal{I}^K$ containing K items. Given a user-item pair (u, i) , the recommender assigns a predicted preference score,

based on which the items are ordered [34, 58]. Let $\mathbf{x} \in \mathbb{R}^{n \times m}$ be a matrix, where $x_{i,j}$ denotes the element of i -th row and j -th column. We define a binary variable $x_{u,i}$ that indicates whether item i appears in $L_K(u)$: specifically, $x_{u,i} = 1$ if item i is included in the list, and $x_{u,i} = 0$ otherwise. The utility of an item i is given by $v_i = \sum_{u \in \mathcal{U}} w_{u,i} x_{u,i}$, which aggregates its utility over all users. Here, $w_{u,i}$ denotes the utility value item i receives from user u , which can be based on various definitions such as exposure, click-through-rate (CTR), or other behavioral signals. Here we utilize the predicted preference score as the value of $w_{u,i} \in [0, 1]$.

Diverse re-ranking. Next we formalize diverse re-ranking. In this paper, we are targeting *aggregate diversity* in the re-ranking phase, *i.e.*, how many distinct items are covered by the system. An item is counted as covered when its utility exceeds a certain threshold δ . In practice, the threshold is often set as $\delta = \eta \sum_{j \in \mathcal{I}} v_j$, where $\eta \in (0, 1)$ is market share (*e.g.*, $\eta = 0.1$ means the threshold is set as 10% of total utility). Let $C_\delta = \{i \in \mathcal{I} : v_i > \delta\}$ denote the set of covered items and coverage is typically defined as the ratio of covered items, $Cov = |C_\delta|/|\mathcal{I}|$. A diverse re-ranking policy aims to construct a new list $L_K^F(u) \in \mathcal{I}^K$ that preserves accuracy while maximizing coverage.

Fair re-ranking. The goal of fair re-ranking is to construct a new list $L_K^F(u) \in \mathcal{I}^K$ that maintains overall accuracy while minimizing disparities, such that $v_i \approx v_j$ for all $i, j \in \mathcal{I}$.

4 Joint Optimization Objective

In this section, we draw inspiration from redistribution theory in economics, and propose an objective function to optimize diversity and fairness while maintaining accuracy. We explain the diversity and fairness objectives separately, and then discuss some nice properties of our objective function.

4.1 Overall Objective

As illustrated in the Introduction, diversity corresponds to the Sufficientarian principle while fairness corresponds to the Rawlsian principle. Drawing inspiration from this correspondence, we propose an objective function to optimize both diversity and fairness in recommender systems. We design the function to be piecewise, where diversity is optimized in the first stage and fairness becomes the target in the latter stage.

Define $f: [0, 1]^{\mathcal{I}} \rightarrow \mathbb{R}$ to be the objective function for utility vector $\mathbf{v} \in [0, 1]^{\mathcal{I}}$:

$$f(\mathbf{v}) = \sum_{i=1}^{\mathcal{I}} \mathcal{F}(v_i), \quad (1)$$

where $\mathcal{F}: [0, 1] \rightarrow \mathbb{R}$ is the function for the utility $v_i \in [0, 1]$ of a single item i :

$$\mathcal{F}(v_i) = \underbrace{\lambda C_\delta^*(v_i)}_{\text{diversity objective}} + \underbrace{U_\alpha^*(v_i)}_{\text{fairness objective}}, \quad (2)$$

where $C_\delta^*(\cdot)$ and $U_\alpha^*(\cdot)$ are diversity and fairness terms, respectively. $\mathcal{F}$ is a piecewise function divided into diversity stage and fairness stage, as detailed in Sections 4.2 and 4.3. $\delta \in (0, 1]$ is the coverage threshold to measure diversity. It is also the dividing point between two stages. $\alpha \in [0, \infty)$ represents the tax rate, where a higher value results in fairer results. Coefficient λ is to ensure the smoothness of

the function at the dividing point, which we will discuss in Section 4.4.

4.2 Diversity Objective

In this section, we introduce how we design the diversity function. First, we define the diversity objective in the following form:

$$C_\delta^*(v) = \begin{cases} 1 - e^{-v/\delta}, & 0 \leq v < \delta, \\ 1 - e^{-1}, & \delta \leq v \leq 1. \end{cases}$$

which takes item utility v as input, and outputs how much it contributes to diversity with coverage threshold set to δ .

Our diversity function aligns with the Bass diffusion model with external influence [6], which models the growth of production adoptions. This function form incentivizes the surge of item utilities to reach the coverage threshold at the early stage of online recommendation, analogous to the Sufficientarian principle in redistribution economics. It "saturates" and falls into a constant when the item is already covered, which aligns with the definition of aggregate diversity. Concavity exhibits a diminishing return growth pattern and encourages breadth (exposing many items) over depth (heavily exposing a few items).

The uniqueness of the objective. To satisfy a set of axioms in Appendix A.1.1 which commonly hold for diversity, Theorem 1 shows that such an objective is unique in RS.

THEOREM 1. *Let $C : [0, \infty) \rightarrow [0, 1)$ be a C^1 function satisfying: (i) $C(0) = 0$; (ii) $C'(v) \geq 0$; (iii) for all $v, w \geq 0$, the memoryless gains identity $C(v + w) = C(v) + (1 - C(v))C(w)$ holds. Then there exists a constant $k \geq 0$ such that for all $v \geq 0$, $C(v) = 1 - e^{-kv}$.*

In particular, $C(\delta) = 1 - e^{-k\delta}$ is a constant for any threshold $\delta > 0$. For simplicity and w.l.o.g., take $k = 1/\delta$ and then $C(v) = 1 - e^{-v/\delta}$.

4.3 Fairness Objective

In this section, we introduce how we design the fairness function based on the iso-elastic utility function and the Rawlsian principle in redistribution economics.

We design a function $U_\alpha^*(v)$ to pursue fairness in the second stage of recommendation when item utility v exceeds coverage threshold δ . From the perspective of Rawlsian redistribution, fairness in re-ranking objectives imposes taxes on richer items and redistributes to poorer items via taxation [54]. Drawing inspiration from iso-elastic utility function [9], we design the fairness function to be:

$$U_\alpha^*(v) = \begin{cases} U_\alpha(\delta), & 0 \leq v < \delta, \\ U_\alpha(v), & \delta \leq v \leq 1. \end{cases}, \quad U_\alpha(v) = \begin{cases} \frac{\gamma}{1-\alpha} v^{1-\alpha}, & \alpha \neq 1, \\ \gamma \ln v, & \alpha = 1. \end{cases}$$

where $\alpha \in [0, \infty)$ denotes the tax parameter, where larger values correspond to stronger emphasis on fairness. The coefficient γ represents the importance weight of each item, and is commonly set to 1 across all items for simplicity.

The fairness component $U_\alpha(v)$ penalizes items with higher utility and redistributes that utility to those with lower utility. To maintain a constant level of overall social welfare, measured as $\sum_i U_\alpha^*(v_i)$, the utility gained by a low-performing item must be offset by a proportional reduction from a high-performing one. More precisely, for each unit of utility transferred to item j , item i must

give up a tax amount of $\frac{\gamma_i}{\gamma_j} (\frac{v_i}{v_j})^\alpha$, ensuring the fairness-preserving transformation.

When $\alpha = 0$, the objective simplifies to a utility-weighted sum, aligning with Utilitarian principles that prioritize efficiency. Setting $\alpha = 1$ leads to the Nash solution, which equalizes the ratio of item utilities to their weights, i.e., $v_i : v_j = \gamma_i : \gamma_j, \forall i, j \in \mathcal{I}$. As α approaches infinity, the objective function prioritizes the least advantaged item, converging to a max-min fairness criterion of the form $\min_{i \in \mathcal{I}} v_i$, thus promoting uniform utility distribution.

4.4 Properties of Objective Function

After introducing the fairness and diversity objectives, we analyze the properties of the overall objective function (shown in Equation 1).

In redistribution economics, smoothness of economic transition is crucial when it comes to the turning point where the leading principle switches from Sufficientarian to Rawlsian. Likewise, to make our piecewise objective function smooth, it is essential to ensure a continuous function value and derivative value between two stages.

To ensure the smoothness of our objective function, we have the following theorem.

THEOREM 2 (PROPERTIES OF FUNCTION f). *The objective function f is L -smooth and μ -strongly concave with*

$$L = \gamma \delta^{-(\alpha+1)} \max\{e, \alpha\}, \quad \mu = \gamma \min\{\alpha, \delta^{-(\alpha+1)}\},$$

where $\lambda (C_\delta^*)'(\delta^-) = (U_\alpha^*)'(\delta^+)$, which leads to $\lambda = \gamma e \delta^{1-\alpha}$.

5 Our Approach: DivFair

Building on the design of the objective function, we propose a re-ranking algorithm, DivFair, which aims to efficiently optimize the objective proposed in Section 4.1. To achieve this, we formulate our multi-objective optimization problem as a mathematical program, and then propose an algorithm called DivFair for online recommendation.

5.1 Optimization Objective

First, the multi-objective re-ranking can be formulated as an integer programming problem. The optimization objective is

$$\mathcal{W}_{\text{OPT}} = \max_{L_K} \sum_{t=1}^T f(L_K(u_t)), \quad \text{s.t.} \quad \mathbf{e} \leq \boldsymbol{\tau}. \quad (3)$$

where f is the objective function based on the ranking list $L_K(u_t)$ for user u_t . $\mathbf{e} \in \mathbb{R}^{\mathcal{I}}$ is the average exposure vector for items across all T users, and $\boldsymbol{\tau} \in \mathbb{R}^{\mathcal{I}}$ denotes the exposure upper limits.

More specifically, based on Equation 1, the diverse and fair re-ranking problem is expressed as

$$\begin{aligned} \mathcal{W}_{\text{OPT}} &= \max_{\mathbf{x}_t \in \mathcal{X}} \sum_{t=1}^T f(\mathbf{x}_t) = \max_{\mathbf{x}_t \in \mathcal{X}} \sum_{t=1}^T \sum_{i=1}^I \frac{1}{\lambda} C_{\delta}^*(\mathbf{v}_{t,i}) + U_{\alpha}^*(\mathbf{v}_{t,i}), \\ \text{s.t. } \mathbf{e} &= \frac{1}{T} \sum_{t=1}^T \mathbf{x}_t \leq \boldsymbol{\tau}, \quad \mathbf{v}_{t,i} = \frac{1}{T} \sum_{t'=1}^t w_{t',i} x_{t',i}, \forall i \in \mathcal{I}, \\ \mathcal{X} &= \left\{ \mathbf{x} \mid x_{t,i} \in \{0, 1\}, \sum_{i \in \mathcal{I}} x_{t,i} = K, \forall t \in [1, 2, \dots, T] \right\}. \end{aligned} \quad (4)$$

where $\mathbf{x}_t \in \{0, 1\}^{|\mathcal{I}|}$ is the decision vector for user u_t . For each item i , $x_{t,i} = 1$ if it is added to the ranking list $L_K(u_t)$, otherwise, $x_{t,i} = 0$. $\mathcal{X}$ is the feasible region of variable $\mathbf{x}$. It ensures that each recommendation list is of size K . The first constraint in Equation 4 suggests that the exposures of each item i are the accumulated exposures over all users. $v_{t,i}$ is utility of item i and here we normalize it to the range $[0, 1]$.

We observe that directly optimizing the Equation 4 is NP-hard since it is a non-linear, large-scale, and integer programming [8]. The time-separable form of the optimization objective inspires us to design an online algorithm to make recommendations to user u_t without access to information beyond time step t . Next we will discuss our proposed model, DivFair, for online recommendation.

5.2 DivFair for Online Recommendation

In this section, we first derive the dual of the original problem, which conveniently turns the primal problem into a convex optimization problem. Then we propose a mirror gradient descent algorithm for efficient online learning with provable error bounds.

5.2.1 Dual problem. From Equation 4 we know that directly solving this integer program offline is NP-hard. However, by deriving its dual problem, we convert the primal problem into a convex optimization problem which is linear with respect to the non-integral dual variable.

THEOREM 3 (DUAL PROBLEM). *The dual problem of Equation 3 can be written as:*

$$\mathcal{W}_{\text{OPT}} \leq \mathcal{W}_{\text{Dual}} = \min_{\boldsymbol{\mu} \in \mathcal{U}} \left[\max_{\mathbf{x}_t \in \mathcal{X}} \left[\sum_{t=1}^T f(\mathbf{x}_t) - \boldsymbol{\mu}^\top \frac{1}{T} \sum_{t=1}^T \mathbf{x}_t \right] + \max_{\mathbf{e} \leq \boldsymbol{\tau}} \boldsymbol{\mu}^\top \mathbf{e} \right], \quad (5)$$

where $\mathcal{U} = \{\boldsymbol{\mu} \mid \mu_i \geq 0, \forall i \in [1, 2, \dots, I]\} = \mathbb{R}_+^I$ is the feasible region of dual variable $\boldsymbol{\mu}$.

Proof of Theorem 3 can be found in Appendix A.3. Since item utility $v_t = v_{t-1} + \frac{1}{T} w_t \odot \mathbf{x}_t$ and $f(\cdot)$ depends on v_t , we write $f(v_t)$ in subsequent sections for brevity.

5.2.2 The DivFair algorithm. Algorithm 1 illustrates DivFair, our proposed method for online optimization. DivFair is designed to operate in the dual space. It leverages linear approximation for real-time decision-making and employs a mirror-descent approach to ensure stable convergence.

DivFair operates on sequences of users of length T . For each user sequence, the algorithm maintains three key components: a dual variable $\boldsymbol{\mu}_t$, a phased budget $\boldsymbol{\theta}_t$ tracking the limits on average exposure, and the accumulated item utility vector $\mathbf{v}_t$. The procedure begins with initialization of $\boldsymbol{\mu}_1 = \mathbf{0}$, $\boldsymbol{\theta}_1 = \mathbf{0}$, and gradient $\mathbf{g}_0 = \mathbf{0}$.

Algorithm 1: DivFair

Input: User arriving order $\{u_t\}_{t=1}^T$, user size T , ranking size K , user-item preference score $w_{t,i}$, $\forall t, i$, and exposure constraints $\boldsymbol{\tau}$.
Output: The decision variables $\{\mathbf{x}_t, i = 1, 2, \dots, N\}$

- 1: Initialize dual solution $\boldsymbol{\mu}_1 = \mathbf{0}$, phased budget $\boldsymbol{\theta}_1 = \mathbf{0}$, item utility $\mathbf{v}_0 = \mathbf{0}$, and gradient $\mathbf{g}_0 = \mathbf{0}$.
- 2: **for** $t = 1, \dots, T$ **do**
- 3: User u_t arrives.
- 4: $\mathbf{m}_t = \begin{cases} 0, & \boldsymbol{\theta}_t \leq \boldsymbol{\tau} \\ \infty, & \text{otherwise} \end{cases}$
- 5: // Online recommendation
- 6: $\mathbf{x}_t = \arg \max_{\mathbf{x}_t \in \mathcal{X}} [f(\mathbf{v}_t) - (\boldsymbol{\mu}_t + \mathbf{m})^\top \mathbf{x}_t / T]$
- 7: $\approx \arg \max_{\mathbf{x}_t \in \mathcal{X}} [f(\mathbf{v}_{t-1}) + \nabla f(\mathbf{v}_{t-1})^\top (\mathbf{v}_t - \mathbf{v}_{t-1}) - (\boldsymbol{\mu}_t + \mathbf{m})^\top \mathbf{x}_t / T]$
- 8: $= \arg \max_{\mathbf{x}_t \in \mathcal{X}} [\nabla f(\mathbf{v}_{t-1})^\top (\mathbf{w}_t \odot \mathbf{x}_t) / T - (\boldsymbol{\mu}_t + \mathbf{m})^\top \mathbf{x}_t / T]$
- 9: $= \arg \max_{\mathbf{x}_t \in \mathcal{X}} [(\nabla f(\mathbf{v}_{t-1}) \odot \mathbf{w}_t - \boldsymbol{\mu}_t - \mathbf{m})^\top \mathbf{x}_t / T]$
- 10: $\boldsymbol{\theta}_{t+1} = \boldsymbol{\theta}_t + \mathbf{x}_t / T$
- 11: $\mathbf{e}_t = \arg \max_{\mathbf{e}_t \leq \boldsymbol{\theta}_t} \boldsymbol{\mu}_t^\top \mathbf{e}_t$
- 12: $\mathbf{v}_t = \mathbf{v}_{t-1} + \mathbf{w}_t \odot \mathbf{x}_t / T$
- 13: $\mathbf{g}_t = \mathbf{e}_t - \mathbf{x}_t / T$
- 14: $\boldsymbol{\mu}_{t+1} = \arg \min_{\boldsymbol{\mu} \in \mathcal{U}} [\langle \mathbf{g}_t, \boldsymbol{\mu} \rangle + \|\boldsymbol{\mu} - \boldsymbol{\mu}_t\|^2 / (2\eta)]$
- 15: **end for**

For each arriving user u_t , the algorithm first computes a penalty term $\mathbf{m}$ to ensure only items whose exposure does not exceed the phased budget are recommended. From line 6 to line 9, we provide a linear approximation of the dual function, through which the scoring function becomes linear with respect to $\mathbf{x}_t$. Therefore, $\mathbf{x}_t$ can be computed using a top- K sorting algorithm with $O(I \log K)$ complexity, suitable for large-scale online recommendation scenarios.

Following the recommendation, we update the phased budget $\boldsymbol{\theta}_t$, accumulated exposure $\mathbf{e}_t$, and the utility vector $\mathbf{v}_t$. Then the subgradient of the dual function is obtained through $\tilde{\mathbf{g}}_t = \mathbf{e}_t - \mathbf{x}_t / T$. Finally, the dual variable is updated through mirror descent. When the gradient $\mathbf{g}_{t,i}$ is positive, the mirror descent step decreases the dual variable, so the algorithm naturally recommends more of this item. With a negative gradient, the value of the dual variable is increased, and the algorithm recommends fewer of this item. Intuitively, in classical welfare economics, the dual variable $\boldsymbol{\mu}$ can be viewed as the shadow price [30], reflecting the cost of allocating additional exposure to an item.

5.3 Theoretical Analysis

In this section, we first provide an upper bound on the linear approximation error. Then we measure the regret of our online algorithm and show that DivFair achieves a sub-linear regret bound. Finally, we compare DivFair with the representative re-ranking model P-MMF, and prove that DivFair ensures a tighter regret bound and more stable convergence.

5.3.1 Linear approximation. Based on Theorem 2, we can bound the error of linear approximation in Algorithm 1.

THEOREM 4 (LINEAR APPROXIMATION). *Let $\mathcal{L}_t(\mathbf{x}_t) := f(\mathbf{v}_t) - (\boldsymbol{\mu}_t + \mathbf{m})^\top \mathbf{x}_t / T$ be the Lagrangian function in line 6 of Algorithm 1. Then in line 7, the error introduced by the linear approximation of $\mathcal{L}(\cdot)$ is of order $O(\frac{1}{T^2})$.*

5.3.2 Regret bound. We measure the regret of an algorithm as the difference between the optimal performance W_{OPT} of the offline problem and the performance of the online algorithm.

$$\text{Regret}(f) = W_{OPT} - W_{online}. \quad (6)$$

We bound the regret in Theorem 5.

THEOREM 5 (REGRET BOUND). *Assume there exists a constant $G \in \mathbb{R}^+$ and $D \in \mathbb{R}^+$ such that $\|g_t\| \leq G$, $\|\mu_1\| \leq D$. Then, the regret can be bounded as follows:*

$$\text{Regret}(f) \leq \frac{D^2}{2\eta} + \frac{\eta G^2 T}{2} + \sum_{t=1}^T \epsilon_t, \quad (7)$$

where $\sum_{t=1}^T \epsilon_t \in \mathcal{O}(\frac{1}{T})$.

Proof of Theorem 5 can be found in Appendix A.5.1. Setting the learning rate as $\eta = \mathcal{O}(\frac{1}{\sqrt{T}})$, we obtain a sublinear regret bound of order $\mathcal{O}(\sqrt{T})$. Then we compare DivFair with the representative re-ranking model P-MMF and provide a corollary in Appendix A.5.2 to show DivFair has a tighter regret bound.

6 Experiments

DivFair is evaluated on three publicly available datasets, and the source code is shared at Github².

6.1 Experimental Settings

Dataset. The experiments are conducted on three large-scale, publicly available ranking datasets, including:

Steam³ [26]: A dataset of user-game interactions from the Steam platform. An interaction is retained only if a user plays a game for more than 10 hours. After pre-processing, it contains 33,832 samples with 4,444 users and 1,237 items.

Amazon_Appliances: A subset of the Amazon Product dataset for appliances⁴. Only clicked data are utilized, simulated as user-item interactions with 4 or 5-star ratings. After pre-processing, it contains 65,836 samples with 8,856 users and 5,515 items.

Yelp⁵: a large-scale business recommendation dataset. Only clicked data are utilized, simulated as user-item interactions with 4 or 5-star ratings. After pre-processing, it contains 260,861 samples with 23,313 users and 15,728 items.

During the pre-processing step, we remove users and items with fewer than five interactions to mitigate sparsity in recommendation data. We sort all interaction data by time and use the first 80% of data to train a base ranking model (i.e., MF ranking model [57]), from which a preference score $w_{u,i}$ is obtained for each user-item pair (u, i) . The remaining 20% of interactions are used as test data for re-ranking.

Evaluation. The performances of the models are evaluated from three aspects: re-ranking accuracy, fairness, and diversity. For accuracy, following [51, 52], we use NDCG@K:

$$\text{NDCG@K} = \frac{1}{|\mathcal{U}|} \sum_{u \in \mathcal{U}} \frac{\sum_{i \in L_K^*(u)} w_{u,i} / \log(\text{rank}_i^* + 1)}{\sum_{i \in L_K(u)} w_{u,i} / \log(\text{rank}_i + 1)}, \quad (8)$$

²<https://github.com/Trailblazers/DivFair>

³http://cseweb.ucsd.edu/~wckang/Steam_games.json.gz

⁴<http://jmcauley.ucsd.edu/data/amazon/>

⁵<https://www.yelp.com/dataset>

where rank_i and rank_i^* denote the positions of the item i in the original ranked list $L_K(u_t)$ and re-ranked list $L_K^*(u_t)$, respectively.

As for fairness, following [18, 19] we utilize Gini Index as our metric, which is defined as:

$$\text{Gini@K} = \frac{\sum_i \sum_j |v_i - v_j|}{2|\mathcal{I}| \sum_i v_i}, \quad (9)$$

where v_i represents the utility of item i . This measure quantifies inequality in utility distribution across items. A score of 0 indicates a perfectly equal distribution, where all items receive identical utility, whereas a score of 1 indicates extreme inequality where a single item captures all the utility while others receive none.

As for diversity performance, we use the widely adopted catalog coverage [21, 22, 28, 49]:

$$\text{Coverage@K} = \frac{|C_v|}{|\mathcal{I}|}. \quad (10)$$

where $C_v = \{i \in \mathcal{I} : v_i > \delta\}$ is the set of covered items whose utility exceeds threshold δ . In this paper, we set the threshold $\delta = \frac{1}{2|\mathcal{I}|} \sum_{i \in \mathcal{I}} v_i$ to be half of the mean utility of all items.

Baselines. We evaluate our method against several representative re-ranking models designed to improve fairness and diversity in RS [53]: **FairRec** [38] and **FairRec+** [10] ensure a max-min share of exposure among items and also evaluate intra-list diversity as part of its evaluation. **Welf** [18] uses Frank-Wolfe optimization routine to jointly optimize a two-sided fairness objective. **CPFair** [36] formulates re-ranking as a knapsack-style problem and applies a greedy algorithm that balances utility and fairness. **P-MMF** [52] employs mirror-descent to improve the utility of the worst-off item group. **Min-regularizer** [52] introduces a regularization term penalizing discrepancies in exposure. **Tax-Rank** [54] constructs a fair re-ranking mechanism through optimal transport, improving coverage by spreading exposure more evenly. **ElasticRank** [55] introduces an economic elasticity-based re-ranking framework to analyze the trade-off between fairness and accuracy, and adjusts distances between item groups to spread coverage more broadly.

Implementation details. All experiments are conducted using Python 3.10 on a server with Ubuntu 20.04. As for the hyperparameters, the learning rate is tuned among $[1e-2, 1e-4]/T^{1/2}$ where T denotes the total number of users. The tax base α is tuned among $[0.0, 10.0]$, and the coverage threshold is varied within $[0.1, 1.0]$. The item weight γ_i is fixed at 1 for all items, and the normalized exposure constraint τ is set to 0.9. We use the most commonly used MF [57] as the ranking base model.

6.2 Experimental Results

In this section, we evaluate the proposed DivFair model and baselines across all three datasets. Three metrics are reported at each ranking size: NDCG (accuracy), Gini Index (fairness), and Coverage (diversity). To ensure a fair comparison, ranking accuracy (measured by NDCG@K) is maintained close to 98%, with an accuracy loss within 1% considered acceptable. This setup reflects a realistic deployment scenario where small decreases in accuracy are permissible if they lead to substantial gains in other system objectives such as fairness and diversity.

Table 1 summarizes the experimental results for DivFair and the baseline models across all datasets with ranking size $K = 10$ and

Table 1: Performance comparison on Steam, Amazon, and Yelp datasets with ranking size $K = 10$ and $K = 20$. The best results in Gini Index and Coverage with NDCG above 97% are shown in boldface, and the second-best are underlined. $\uparrow$ and $\downarrow$ indicate whether a higher or lower value is better, respectively.

Top-K	Models	K=10			K=20		
		NDCG@10 $\uparrow$	GINI@10 $\downarrow$	Cov@10 $\uparrow$	NDCG@20 $\uparrow$	GINI@20 $\downarrow$	Cov@20 $\uparrow$
Steam	FairRec	0.9529	0.3309	0.7851	0.9496	0.3330	0.8222
	FairRec+	0.9531	0.3313	0.7779	0.9497	0.3328	0.8223
	TaxRank	0.9472	0.3056	0.7900	0.9793	0.4820	0.6236
	CPFair	0.9912	0.3534	0.6906	0.9850	0.3776	0.6333
	Min-Regularizer	0.9737	0.3020	0.7714	0.9729	0.3266	0.6955
	Welf	0.9813	<u>0.3007</u>	<u>0.7827</u>	0.9779	0.3407	<u>0.7439</u>
	ElasticRank	0.9848	0.3583	0.6914	0.9922	0.4505	0.5985
	P-MMF	0.9805	0.3154	0.7342	0.9838	0.3722	0.6284
	DivFair (Ours)	0.9834	0.2648	0.8570	0.9796	0.3051	0.8086
Amazon	FairRec	0.9562	0.2863	0.5611	0.9544	0.2292	0.7980
	FairRec+	0.9579	0.2967	0.5393	0.9549	0.2336	0.7736
	TaxRank	0.9432	0.2791	0.7030	0.9457	0.2529	0.7320
	CPFair	0.9857	0.3274	0.4185	0.9875	0.3228	0.4053
	Min-Regularizer	0.9846	0.3381	0.5063	0.9828	<u>0.2829</u>	0.5069
	Welf	0.9711	0.3400	<u>0.5633</u>	0.9809	0.3244	0.5886
	ElasticRank	0.9882	0.3990	0.2139	0.9900	0.3800	0.2673
	P-MMF	0.9834	<u>0.3219</u>	0.5063	0.9849	0.3151	0.5063
	DivFair (Ours)	0.9805	0.2480	0.7762	0.9844	0.2406	0.7591
Yelp	FairRec	0.9664	0.2851	0.6158	0.9660	0.2607	0.7229
	FairRec+	0.9681	0.2902	0.6113	0.9666	0.2586	0.7252
	TaxRank	0.9545	0.2517	0.7095	0.9555	0.2294	0.7392
	CPFair	0.9882	0.1889	0.4919	0.9896	0.1846	0.4975
	Min-Regularizer	0.9715	<u>0.0820</u>	0.5328	0.9724	0.0730	<u>0.5328</u>
	Welf	0.9789	0.5409	0.4427	0.9869	0.5061	0.4636
	ElasticRank	0.9893	0.2104	0.4072	0.9913	0.2099	0.3926
	P-MMF	0.9887	0.2024	0.4868	0.9897	0.1992	0.4913
	DivFair (Ours)	0.9847	0.0763	0.8710	0.9862	<u>0.0737</u>	0.9136

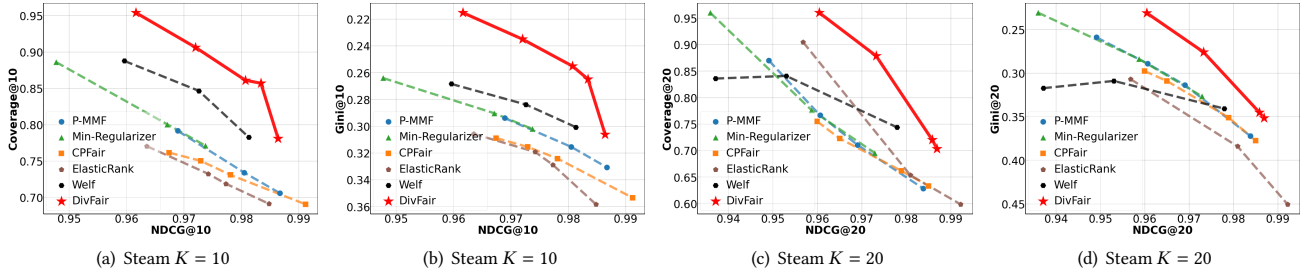


Figure 2: Pareto frontier on the Steam dataset with different ranking size K .

$K = 20$. We tune different re-ranking models to achieve similar accuracy (NDCG close to 98%) and evaluate them in terms of fairness and diversity. We observe that FairRec, FairRec+, and TaxRank show limited ability to jointly preserve accuracy and improve fairness or diversity. These methods either sacrifice accuracy for fairness or achieve only modest fairness gains. By contrast, DivFair consistently achieves superior fairness and diversity outcomes under comparable accuracy levels, demonstrating the effectiveness of our approach.

To further illustrate the trade-offs among accuracy, fairness, and diversity, Figure 2 presents the Pareto frontiers on the Steam dataset. We only compare the best trade-off baselines, excluding FairRec, FairRec+, and TaxRank. The frontiers are constructed by tuning the hyperparameters of each method and plotting the best-performing points. From these results, DivFair clearly dominates the baselines: under the same NDCG@K level, DivFair achieves higher coverage

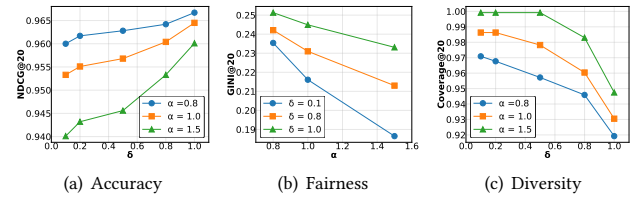


Figure 3: Accuracy, fairness and diversity change w.r.t. key parameters: the tax rate α and coverage threshold δ .

and lower Gini Index, while under the same Coverage or Gini Index, DivFair attains higher NDCG@K. These results confirm that DivFair effectively distributes utility both equitably and broadly across the item catalog, validating its superiority in jointly optimizing fairness and diversity.

Table 2: Performance comparison on Steam with different base models with ranking size $K = 20$. The best results in Gini Index and Coverage are shown in boldface, and the second-best are underlined.

Base Model	Top-K Re-ranking Model	K=20		
		NDCG $\uparrow$	GINI $\downarrow$	Cov $\uparrow$
BPR	P-MMF	0.9876	0.1343	0.7542
	CPFair	0.9899	0.1898	0.6370
	ElasticRank	0.9898	0.1990	0.5845
	DivFair (Ours)	0.9873	0.1054	0.9709
GRU4Rec	P-MMF	0.9825	0.7014	0.5028
	CPFair	0.9733	<u>0.6590</u>	<u>0.5546</u>
	ElasticRank	0.9784	0.7842	0.2797
	DivFair (Ours)	0.9748	0.5204	0.7219

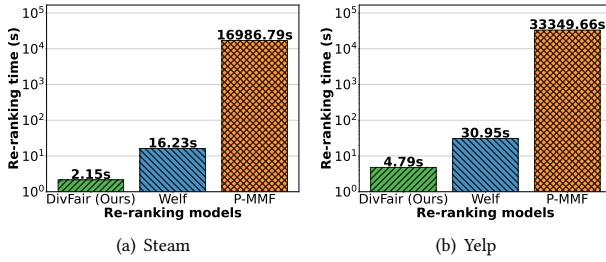


Figure 4: Inference time comparison between DivFair and two best-performing baselines.

6.3 Experiment Analysis

We also conduct experiments to analyze DivFair under $K = 20$. Similar results can be observed in other datasets and other K values.

6.3.1 Performance with different base models. Since DivFair and the baseline methods are re-ranking models, they operate on the ranking lists generated by an underlying base ranking model. To assess the robustness of DivFair under different base ranking models, we experiment with another two widely used architectures for generating user-item preference scores, including **BPR** [41], a pairwise ranking model that learns personalized rankings from implicit feedback by optimizing relative preferences; and **GRU4Rec** [44], a sequence-based recurrent neural network model that captures temporal or session-level dependencies in user behavior. All experiments are conducted on Steam, and similar trends can be observed in the other two datasets.

Table 2 reports the performance of DivFair compared to three best-performing re-ranking baselines under these base models. The results show that DivFair consistently achieves superior fairness and diversity while maintaining high accuracy, regardless of the underlying base recommender. This confirms the robustness and generalizability of DivFair in re-ranking outputs from different base models.

6.3.2 Ablation study on different parameters. We conduct an ablation study on two key hyperparameters of DivFair: the tax rate α and the coverage threshold δ . α ranges from $[0.8, 1.0, 1.5]$ while δ takes values from $[0.1, 0.2, 0.5, 0.8, 1.0]$. Figure 3 presents how these parameters affect ranking accuracy (NDCG), fairness (Gini Index), and diversity (Coverage) on the Steam dataset with $K = 20$.

From Figure 3(a), we observe that larger δ and smaller α prioritize accuracy and improve NDCG. Figure 3(b) shows that Gini Index is lower with larger α , indicating better fairness. In Figure 3(c), Coverage decreases as δ increases, but this effect is mitigated when α is large, since stronger fairness constraints boost the utilities of long-tailed items.

These trends can be explained by the curvature properties of the fairness and diversity functions. The fairness utility function in Section 4.3 becomes more concave with sharper curvature as α increases, imposing stronger penalties on high-utility items and redistributing utility more evenly, thereby improving fairness but reducing accuracy. Meanwhile, the diversity function in Section 4.2 rises more sharply for smaller δ , enabling more items to meet the coverage threshold quickly. Together, larger α and smaller δ lead to a more concave objective, favoring fairness and diversity at the cost of accuracy. In contrast, smaller α and larger δ flatten the gradients, yielding higher ranking accuracy.

6.3.3 Inference time. To assess the efficiency of our approach, we compare the inference time of DivFair on the Steam and Yelp dataset against two baseline re-ranking models, Welf and P-MMF, which demonstrate competitive performance in Section 6.2. We adapt P-MMF to individual-level fairness and report the results in Figure 4. The x-axis represents re-ranking models, and the y-axis shows the total re-ranking time across all users in log scale.

As shown in Figure 4, DivFair achieves significantly faster inference compared to the baselines. Specifically, DivFair is up to seven times faster than Welf and nearly four orders of magnitude faster than P-MMF. This efficiency stems from our design in the DivFair Algorithm: instead of solving complex non-linear optimization problems, DivFair leverages linear approximation in dual space that maintains the same time complexity as standard sorting algorithms.

These results show that DivFair is not only effective but also efficient for large-scale deployment in online RS where inference speed is critical.

7 Conclusion

In this paper, we propose a novel re-ranking model, DivFair, to unify fairness and diversity in recommender systems by drawing on principles from economic growth theory. We design an objective function that improves diversity via the Sufficentarian principle and fairness via the Rawlsian principle. To optimize this objective in online settings, we develop a mirror descent algorithm, employing linear approximation to make online recommendations efficiently. Theoretical analysis shows that DivFair achieves sub-linear regret and stable convergence. Experiments on three large-scale datasets demonstrate that DivFair outperforms state-of-the-art baselines in both re-ranking effectiveness and computational efficiency.

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References

- [1] Himan Abdollahpour, Masoud Mansoury, Robin Burke, and Bamshad Mobasher. 2019. The Unfairness of Popularity Bias in Recommendation. *arXiv preprint arXiv:1907.13286* (2019).
- [2] Irma Adelman. 1961. *Theories of Economic Growth and Development*. Stanford University Press, Stanford, CA.
- [3] Gediminas Adomavicius and YoungOk Kwon. 2011. Improving the recommender system performance through diversity and neighborhood selection. *INFORMS Journal on Computing* 23, 4 (2011), 574–592.
- [4] Gediminas Adomavicius and YoungOk Kwon. 2012. Improving aggregate recommendation diversity using ranking-based techniques. In *Proceedings of the Workshop on Recommendation Utility Evaluation: Beyond RMSE (RUE)*.
- [5] C. Barry. 2004. Redistribution. Stanford Encyclopedia of Philosophy. <https://plato.stanford.edu/entries/redistribution/> Online; retrieved 13 August 2010.
- [6] Frank M. Bass. 1969. A new product growth for model consumer durables. *Management Science* 15, 5 (1969), 215–227.
- [7] W.J. Baumol and A.S. Blinder. 2011. *Microeconomics: Principles and Policy*. Cengage Learning. <https://books.google.com.sg/books?id=RFVrcK4UW7EC>
- [8] Dimitri P Bertsekas. 1997. Nonlinear programming. *Journal of the Operational Research Society* 48, 3 (1997), 334–334.
- [9] Dimitris Bertsimas, Vivek F. Farias, and Nikolaos Trichakis. 2012. On the Efficiency-Fairness Trade-off. *Management Science* 58, 12 (2012), 2234–2250.
- [10] Arpita Biswas, Gourab K. Patro, Niloy Ganguly, Krishna P Gummadi, and Abhijnan Chakraborty. 2021. Toward Fair Recommendation in Two-sided Platforms. *ACM Transactions on the Web (TWEB)* 16, 2 (2021), 1–34.
- [11] Stephen P. Boyd and Lieven Vandenbergh. 2004. *Convex Optimization*. Cambridge University Press, Cambridge. 216 pages. <https://web.stanford.edu/~boyd/cvxbook/> Retrieved October 15, 2011.
- [12] Jaime Carbonell and Jade Goldstein. 1998. The use of MMR, diversity-based reranking for reordering documents and producing summaries. In *Proceedings of the 21st annual international ACM SIGIR conference on Research and development in information retrieval*. ACM, 335–336.
- [13] Nicolò Cesa-Bianchi, Claudio Gentile, Gábor Lugosi, and Gergely Neu. 2017. Boltzmann Exploration Done Right. *arXiv:1705.10257 [cs.LG]* <https://arxiv.org/abs/1705.10257>
- [14] Junnan Chen, Hongzhi Zhang, Yongfeng Zhang, Jialin Ma, Wayne Xin Liu, and Shaoping Ma. 2018. Fast greedy MAP inference for determinantal point process to improve recommendation diversity. In *Proceedings of the 32nd AAAI Conference on Artificial Intelligence*. 3313–3320.
- [15] Minmin Chen, Alex Beutel, Paul Covington, Sagar Jain, Francois Belletti, and Ed Chi. 2018. Top-K Off-Policy Correction for a REINFORCE Recommender System. *arXiv preprint arXiv:1812.02353* (2018). <https://arxiv.org/abs/1812.02353>
- [16] John Bates Clark. [n.d.]. *The Distribution of Wealth: A Theory of Wages, Interest and Profits*. Macmillan. https://books.google.com/books/about/The_Distribution_of_Wealth.html?id=laM4zQEACAAJ Analytical Table of Contents edition (1902).
- [17] Yashar Deldjoo, Dietmar Jannach, Alejandro Bellogin, Alessandro Difonzo, and Dario Zanzonelli. 2022. A Survey of Research on Fair Recommender Systems. *arXiv preprint arXiv:2205.11127* (2022).
- [18] Virginie Do, Sam Corbett-Davies, Jamal Atif, and Nicolas Usunier. 2021. Two-sided Fairness in Rankings via Lorenz Dominance. *Advances in Neural Information Processing Systems* 34 (2021), 8596–8608.
- [19] Virginie Do and Nicolas Usunier. 2022. Optimizing Generalized Gini Indices for Fairness in Rankings. In *Proceedings of the 45th International ACM SIGIR Conference on Research and Development in Information Retrieval*. 737–747.
- [20] David A. Freedman. 1975. On Tail Probabilities for Martingales. *The Annals of Probability* 3, 1 (1975), 100–118.
- [21] Mouzhi Ge, Carla Delgado-Battenfeld, and Dietmar Jannach. 2010. Beyond Accuracy: Evaluating Recommender Systems by Coverage and Serendipity. In *Proceedings of the 4th ACM Conference on Recommender Systems (RecSys)*. 257–260.
- [22] Qingyao Hong, Yifeng Zhao, Yongfeng Zhang, and Yongfeng Wu. 2025. Multi-Objective Generative Recommendation: A Survey of Beyond-Accuracy Objectives and Techniques. *arXiv preprint arXiv:2404.11334* (2025). <https://arxiv.org/abs/2404.11334>
- [23] Neil Hurley. 2016. Personalised ranking with diversity. *Information Retrieval Journal* 19, 3 (2016), 313–337.
- [24] Robert Huseby. 2019. Sufficientarianism. *Oxford Research Encyclopedia of Politics* (2019). Published online 29 July 2019.
- [25] Marius Kaminskis and Derek Bridge. 2016. Diversity, Serendipity, Novelty, and Coverage: A Survey and Empirical Analysis of Beyond-Accuracy Objectives in Recommender Systems. *ACM Transactions on Interactive Intelligent Systems (TiiS)* 7, 1 (2016), 1–42.
- [26] Wang-Cheng Kang and Julian McAuley. 2018. Self-attentive Sequential Recommendation. In *2018 IEEE international conference on data mining (ICDM)*. IEEE, 197–206.
- [27] Wouter Kool, Herke van Hoof, and Max Welling. 2019. Stochastic Beams and Where To Find Them: The Gumbel-Top-k Trick for Sampling Sequences Without Replacement. In *Proceedings of the 36th International Conference on Machine Learning (ICML)*. 3499–3508. <https://proceedings.mlr.press/v97/kool19a.html>
- [28] Iordanis Koutsopoulos and Maria Halkidi. 2018. Efficient and Fair Item Coverage in Recommender Systems. In *DASC / PiCom / DataCom / CyberSciTech (IEEE)*. 912–918.
- [29] Matevz Kunaver and Tomaž Požrl. 2017. Diversity in Recommender Systems – A Survey. *Knowledge-Based Systems* 123 (2017), 154–162.
- [30] Peter J. Lambert. 1992. *The Distribution and Redistribution of Income*. Springer.
- [31] Yunqi Li, Hanxiong Chen, Zuohui Fu, Yingqiang Ge, and Yongfeng Zhang. 2021. User-oriented Fairness in Recommendation. In *Proceedings of the Web Conference*. 624–632.
- [32] Yunqi Li, Hanxiong Chen, Shuyuan Xu, Yingqiang Ge, Juntao Tan, Shuchang Liu, and Yongfeng Zhang. 2022. Fairness in Recommendation: A Survey. *arXiv preprint arXiv:2205.13619* (2022).
- [33] Yunqi Li, Hanxiong Chen, Shuyuan Xu, Yingqiang Ge, Juntao Tan, Shuchang Liu, and Yongfeng Zhang. 2023. Fairness in Recommendation: Foundations, Methods and Applications. *ACM Transactions on Intelligent Systems and Technology* 14, 5 (2023), Article No.: 95.
- [34] Xiangyu Liu, Chuan Yu, Zhilin Zhang, Zhenzhe Zheng, Yu Rong, Hongtao Lv, Da Huo, Yiqing Wang, Dagui Chen, Jian Xu, et al. 2021. Neural Auction: End-to-end Learning of Auction Mechanisms for E-commerce Advertising. In *Proceedings of the 27th ACM Conference on Knowledge Discovery & Data Mining*. 3354–3364.
- [35] Lars Ljungqvist and Thomas J. Sargent. 2000. *Recursive Macroeconomic Theory*. MIT Press, London. 451 pages.
- [36] Mohammadmehdi Naghiaei, Hossein A Rahmani, and Yashar Deldjoo. 2022. CP-Fair: Personalized Consumer and Producer Fairness Re-ranking for Recommender Systems. *arXiv preprint arXiv:2204.08085* (2022).
- [37] Arkadi Nemirovsky and David Yudin. 1983. *Problem Complexity and Method Efficiency in Optimization*. John Wiley & Sons, New York.
- [38] Gourab K. Patro, Arpita Biswas, Niloy Ganguly, Krishna P. Gummadi, and Abhijnan Chakraborty. 2020. FairRec: Two-sided Fairness for Personalized Recommendations in Two-sided Platforms. In *Proceedings of the Web Conference*. 1194–1204.
- [39] Shameem Puthiya Parambath, Siwei Liu, Christos Anagnostopoulos, Roderick Murray-Smith, and Iadh Ounis. 2022. Parameter Tuning of Reranking-based Diversification Algorithms using Total Curvature Analysis. In *Proceedings of the 2022 ACM SIGIR International Conference on Theory of Information Retrieval*. 108–114.
- [40] Martin Ravallion. 2008. Poverty Lines. In *The New Palgrave Dictionary of Economics, 2nd Edition*, Larry Blume and Steven Durlauf (Eds.). Palgrave Macmillan, London. https://www.uni-trier.de/fileadmin/fb4/prof/VWL/TWB/Vorlesung/neu/Ravallion_-_Poverty_Lines.pdf “Analytical Table of Contents” version.
- [41] Steffen Rendle, Christoph Freudenthaler, Zeno Gantner, and Lars Schmidt-Thieme. 2012. BPR: Bayesian Personalized Ranking from Implicit Feedback. *arXiv preprint arXiv:1205.2618* (2012).
- [42] Prasanta Sahoo, Sasmita Khandual, and Susanta Sahoo. 2025. Rawlsian justice and welfare state: Foundations, critiques and institutional alternatives. *International Journal of Political Science and Governance* 7, 7, Part B (2025), 115–120.
- [43] Ashudeep Singh and Thorsten Joachims. 2019. Policy Learning for Fairness in Ranking. *Advances in Neural Information Processing Systems* 32 (2019).
- [44] Yong Kiam Tan, Xinxing Xu, and Yong Liu. 2016. Improved Recurrent Neural Networks for Session-based Recommendations. In *Proceedings of the 1st Workshop on Deep Learning for Recommender Systems (DLRS)*. ACM Press, 17–22.
- [45] Bertil Tungodden and Peter Vallentyne. 2006. Who Are the Least Advantaged? In *Egalitarianism: New Essays on the Nature and Value of Equality*, Nils Holtug and Kasper Lippert-Rasmussen (Eds.). Oxford University Press, Oxford, 174–195.
- [46] Yifan Wang, Weizhi Ma, Min Zhang, Yiqun Liu, and Shaoping Ma. 2022. A Survey on the Fairness of Recommender Systems. *ACM Transactions on Information Systems* 40, 4 (2022), 1–37.
- [47] Mesfin Mulugeta Woldegiorgis. 2022. Inequality, Social Protection Policy, and Inclusion: Pertinent Theories and Empirical Evidence. *Journal of Social and Economic Development* 24, 2 (2022), 241–265.
- [48] Chen Wu, Yifeng Zhao, Yongfeng Zhang, and Yongfeng Wu. 2024. LLM4Rerank: Reranking with Large Language Models. In *Proceedings of the 47th International ACM SIGIR Conference on Research and Development in Information Retrieval (SIGIR)*.
- [49] Haolun Wu, Yansen Zhang, Chen Ma, Fuyuan Lyu, Bowei He, Bhaskar Mitra, and Xue Liu. 2024. Result Diversification in Search and Recommendation: A Survey. *IEEE Trans. on Knowl. and Data Eng.* 36, 10 (Oct. 2024), 5354–5373.
- [50] Le Wu, Qi Liu, Enhong Chen, Nicholas Jing Yuan, Guangming Guo, and Xing Xie. 2016. Relevance meets coverage: A unified framework to generate diversified recommendations. *ACM Transactions on Intelligent Systems and Technology (TIST)* 7, 3 (2016), 1–30.
- [51] Yao Wu, Jian Cao, Guandong Xu, and Yudong Tan. 2021. TFROM: A Two-sided Fairness-aware Recommendation Model for both Customers and Providers. In *Proceedings of the 44th International ACM SIGIR Conference on Research and Development in Information Retrieval*. 1013–1022.
- [52] Chen Xu, Sirui Chen, Jun Xu, Weiran Shen, Xiao Zhang, Gang Wang, and Zhenhua Dong. 2023. P-MMF: Provider Max-min Fairness Re-ranking in Recommender

- System. In *Proceedings of the ACM Web Conference 2023*. 3701–3711.
- [53] Chen Xu, Zhirui Deng, Clara Rus, Xiaopeng Ye, Yuanna Liu, Jun Xu, Zhicheng Dou, Ji-Rong Wen, and Maarten de Rijke. 2025. FairDiverse: A Comprehensive Toolkit for Fairness- and Diversity-aware Information Retrieval. In *Proceedings of the 48th International ACM SIGIR Conference on Research and Development in Information Retrieval* (Padua, Italy) (SIGIR '25). Association for Computing Machinery, New York, NY, USA, 3540–3550.
- [54] Chen Xu, Xiaopeng Ye, Wenjie Wang, Liang Pang, Jun Xu, and Tat-Seng Chua. 2024. A Taxation Perspective for Fair Re-ranking. In *Proceedings of the 47th International ACM SIGIR Conference on Research and Development in Information Retrieval* (Washington DC, USA) (SIGIR '24). Association for Computing Machinery, New York, NY, USA, 1494–1503.
- [55] Chen Xu, Jujia Zhao, Wenjie Wang, Liang Pang, Jun Xu, Tat-Seng Chua, and Maarten de Rijke. 2025. Understanding Accuracy-Fairness Trade-offs in Re-ranking through Elasticity in Economics. In *Proceedings of the 48th International ACM SIGIR Conference on Research and Development in Information Retrieval*. 539–548.
- [56] Jun Xu, Xiangnan He, and Hang Li. 2018. Deep Learning for Matching in Search and Recommendation. In *The 41st International ACM SIGIR Conference on Research & Development in Information Retrieval* (Ann Arbor, MI, USA) (SIGIR '18). Association for Computing Machinery, New York, NY, USA, 1365–1368.
- [57] Hong-Jian Xue, Xinyu Dai, Jianbing Zhang, Shujian Huang, and Jiajun Chen. 2017. Deep Matrix Factorization Models for Recommender Systems.. In *IJCAI*, Vol. 17. Melbourne, Australia, 3203–3209.
- [58] Xun Yang, Yasong Li, Hao Wang, Di Wu, Qing Tan, Jian Xu, and Kun Gai. 2019. Bid Optimization by Multivariable Control in Display Advertising. In *Proceedings of the 25th ACM international conference on knowledge discovery & data mining*. 1966–1974.
- [59] W. H. Young. 1912. On classes of summable functions and their Fourier series. *Proceedings of the Royal Society A* 87, 594 (1912), 225–229.
- [60] Yifeng Zhao, Yifan Li, Xiangyu Li, Yujie Xian, Yongfeng Zhang, and Yongfeng Wu. 2024. InFoDance: Influence-guided Data-centric Optimization for Fair and Diverse Recommendations. In *Proceedings of the 47th International ACM SIGIR Conference on Research and Development in Information Retrieval* (SIGIR).
- [61] Yifeng Zhao and Yongfeng Wu. 2024. A Survey on Fairness and Diversity in Recommender Systems: Concepts, Metrics, and Challenges. *arXiv preprint arXiv:2402.07746* (2024). <https://arxiv.org/abs/2402.07746>
- [62] Cai-Nicolas Ziegler, Sean M McNee, Joseph A Konstan, and Georg Lausen. 2005. Improving recommendation lists through topic diversification. In *Proceedings of the 14th international conference on World Wide Web*. ACM, 22–32.

A Appendix

A.1 The Uniqueness of Diversity Objective

A.1.1 Diversity axioms in RS. Based on the concept of aggregate diversity and a rich line of related work [39, 50, 62], the diversity function $C_\delta(\cdot)$ should satisfy the following axioms.

- A1 (Monotonicity)** $C'_\delta(v) \geq 0, \forall v \geq 0$. More utility never reduces diversity.
- A2 (Law of diminishing returns)** $C''_\delta(v) \leq 0, \forall v \geq 0$. The marginal gain in diversity decreases as accumulated utility grows. Diminishing return is a well-understood fundamental principle of economics [7].
- A3 (Capped potential)** $C_\delta(0) = 0, \lim_{v \rightarrow \infty} C_\delta(v) = 1$. Diversity is zero before any exposure happens, and has an upper ceiling of 1.
- A4 (Memoryless gains)** For all $v, w \geq 0, C_\delta(v+w) = C_\delta(v) + (1 - C_\delta(v))\psi(w)$, where $\psi(w)$ does not depend on v . A fixed extra utility w converts the *remaining* potential into diversity by the *same factor*, regardless of history [39, 50]. Mathematically, there exists a function $\psi : [0, \infty) \rightarrow [0, 1]$ independent of v such that the gain from adding w to accumulated utility v satisfies a consistent multiplicative structure.
- A5 (Threshold-invariant)** $\exists C_0 \in (0, 1)$ such that $C_\delta(\delta) = C_0, \forall \delta > 0$. C_0 is the same constant for all δ , independent of

the threshold value.

A.1.2 Proof sketch of Theorem 1. Due to space limit, here we provide a sketch of proof including all major steps. The complete proof is shared at <https://github.com/Trailblazerss/DivFair>.

PROOF. Let the coverage level after accumulated utility v be the scalar $X(v) = C_\delta(v) \in [0, 1]$. Observables are bounded Borel functions $f : [0, 1] \rightarrow \mathbb{R}$ in $\mathcal{B} = \{f : \|f\|_\infty < \infty\}$, the Banach space under the sup norm $\|\cdot\|_\infty$.

For every budget chunk $w \geq 0$, define an operator

$$(T_w f)(x) = f(x + (1-x)C_\delta(w)), \quad x \in [0, 1]. \quad (1.1)$$

We can show $\{T_w\}_{w \geq 0} \subset \mathcal{L}(\mathcal{B})$ is a contractive C_0 -semigroup, by proving $(T_{w_1} T_{w_2} f)(x) = (T_{w_1+w_2} f)(x)$, and its strong continuity.

By considering its infinitesimal generator and using (1.1), we can show the semigroup satisfies

$$\frac{d}{dv} T_v f = A T_v f, \quad T_0 = I.$$

For $m(v) := T_v f(0) = C_\delta(v)$ we get an ODE

$$\frac{dm}{dv} = -\kappa(1 - m(v)), \quad m(0) = 0,$$

Combined with Axiom A5, the solution leads to $C_\delta(v) = 1 - e^{-v/\delta}$. $\square$

A.2 Proof of Theorem 2

PROOF. From definitions in Section 4, we derive the first and second derivatives of $C_\delta^*(\cdot)$ and $U_\alpha^*(\cdot)$.

$$C_\delta^*{}'(v) = \begin{cases} \frac{e^{-v/\delta}}{\delta}, & 0 \leq v < \delta, \\ 0, & \delta < v \leq 1, \end{cases}, \quad C_\delta^*{}''(v) = \begin{cases} -\frac{e^{-v/\delta}}{\delta^2}, & 0 \leq v < \delta, \\ 0, & \delta < v \leq 1. \end{cases}$$

$$U_\alpha^*{}'(v) = \begin{cases} \gamma v^{-\alpha}, & \alpha \neq 1, \\ \frac{\gamma}{v}, & \alpha = 1, \end{cases}, \quad U_\alpha^*{}''(v) = \begin{cases} -\alpha \gamma v^{-\alpha-1}, & \alpha \neq 1, \\ -\frac{\gamma}{v^2}, & \alpha = 1. \end{cases}$$

Therefore, we have the derivatives of $\mathcal{F}(\cdot)$.

$$\mathcal{F}'(v) = \begin{cases} \frac{1}{\lambda} \frac{e^{-v/\delta}}{\delta}, & 0 \leq v < \delta, \\ \gamma v^{-\alpha}, & \delta < v \leq 1, \end{cases}, \quad \mathcal{F}'(\delta^-) = \mathcal{F}'(\delta^+) = \frac{e^{-1}}{\lambda \delta}.$$

and $\mathcal{F}'_1(v)$ is continuous at the single "kink" point $v = \delta$.

$$\mathcal{F}''(v) = \begin{cases} -\frac{1}{\lambda} \frac{e^{-v/\delta}}{\delta^2}, & 0 \leq v < \delta, \\ -\alpha \gamma v^{-\alpha-1}, & \delta < v \leq 1, \\ \text{undefined (kink)}, & v = \delta. \end{cases}$$

By deriving the maximum and minimum value of $|\mathcal{F}''(v)|$, one can quickly find that $L \triangleq \max_v |\mathcal{F}''(v)| = \gamma \delta^{-(\alpha+1)} \max\{e, \alpha\}$, $\mu \triangleq \min_v |\mathcal{F}''(v)| = \gamma \min\{\alpha, \delta^{-(\alpha+1)}\}$. The zero-measure non-differentiable slice $v_i = \delta$ does not affect the integrals, so the bounds hold on all of $[0, 1]^T$.

Therefore, we show that the objective function f defined in Section 4 is L -smooth and μ -strongly concave. $\square$

A.3 Proof of Theorem 3

PROOF. Since the exposure vector $\mathbf{e} = \frac{1}{T} \sum_{t=1}^T \mathbf{x}_t \leq \boldsymbol{\tau}$, we can treat the $\mathbf{e}$ as an auxiliary variable and move the exposure constraints to the objective using a vector of Lagrange multipliers $\boldsymbol{\mu} \in \mathbb{R}^I$. From minimax inequality, we have

$$\begin{aligned} \mathcal{W}_{\text{OPT}} &= \max_{\mathbf{x}_t \in \mathcal{X}} \min_{\mathbf{e} \leq \boldsymbol{\tau}} \min_{\boldsymbol{\mu} \in \mathcal{U}} \left[\sum_{t=1}^T f(\mathbf{x}_t) - \boldsymbol{\mu}^\top \left(-\mathbf{e} + \frac{1}{T} \sum_{t=1}^T \mathbf{x}_t \right) \right] \\ &\leq \min_{\boldsymbol{\mu} \in \mathcal{U}} \left[\max_{\mathbf{x}_t \in \mathcal{X}} \left[\sum_{t=1}^T f(\mathbf{x}_t) - \boldsymbol{\mu}^\top \frac{1}{T} \sum_{t=1}^T \mathbf{x}_t \right] + \max_{\mathbf{e} \leq \boldsymbol{\tau}} \boldsymbol{\mu}^\top \mathbf{e} \right] = \mathcal{W}_{\text{Dual}} \end{aligned}$$

where $\mathcal{U}$ is the feasible region of dual variable $\boldsymbol{\mu}$. To make the conjugate function $r^*(\boldsymbol{\mu}) = \max_{\mathbf{e} \leq \boldsymbol{\tau}} \boldsymbol{\mu}^\top \mathbf{e}$ finite, it suffices to have non-negative vector $\boldsymbol{\mu}$. So $\mathcal{U} = \{\boldsymbol{\mu} | \mu_i \geq 0, \forall i \in [1, 2, \dots, I]\} = \mathbb{R}_+^I$. $\mathcal{U}$ is a closed and convex set with positive orthant inside its recession cone. $\square$

A.4 Proof of Linear Approximation Error

A.4.1 Taylor expansion. We first provide a lemma to show the error between objective function and its first-order Taylor expansion.

LEMMA 1 (TAYLOR EXPANSION OF f). *If $\Delta_f(v_t)$ denotes the error of using first-order Taylor expansion to approximate the objective function $f(\cdot)$, then $\Delta_f(v_t) \in O(\frac{1}{T^2})$.*

PROOF. Let $\Delta_f(v_t) := \tilde{f}(v_t) - f(v_t)$, and $\tilde{f}(v_t) := f(v_{t-1}) + \nabla f(v_{t-1})^\top (v_t - v_{t-1})$.

From the proof of Theorem 2 we know that function $f: [0, 1]^I \rightarrow \mathbb{R}$ is L-smooth with $\Delta_f(v) \geq 0$. So the first derivative of f is absolutely continuous on the closed interval between 0 and 1.

Let $\mathbf{z}_t := v_t - v_{t-1} = \frac{1}{T} \mathbf{w}_t \odot \mathbf{x}_t$. According to Taylor's theorem, we have the integral form of the remainder that

$$\begin{aligned} \Delta_f(v_t) &= |\tilde{f}(v_t) - f(v_t)| = |f(v_{t-1}) + \nabla f(v_{t-1})^\top \mathbf{z}_t - f(v_{t-1} + \mathbf{z}_t)| \\ &= \left| \int_{v_{t-1}}^{v_t} f''(t)(v_t - t) dt \right| \leq L \int_{v_{t-1}}^{v_t} (v_t - t) dt = \frac{L}{2} |\mathbf{z}_t^2| \\ &= \frac{L}{2T^2} |\mathbf{w}_t \odot \mathbf{x}_t|^2 \leq \frac{L \mathbf{w}_{\max}^2 k}{2T^2} \in O\left(\frac{1}{T^2}\right). \end{aligned}$$

$\square$

A.4.2 Proof of Theorem 4.

PROOF. Let

$$\mathcal{L}_t(\mathbf{x}_t) := f(\mathbf{v}_t) - (\boldsymbol{\mu}_t + \mathbf{m})^\top \mathbf{x}_t / T,$$

$$\tilde{\mathcal{L}}_t(\mathbf{x}_t) := f(\mathbf{v}_{t-1}) + \nabla f(\mathbf{v}_{t-1})^\top (\mathbf{v}_t - \mathbf{v}_{t-1}) - (\boldsymbol{\mu}_t + \mathbf{m})^\top \mathbf{x}_t / T,$$

and $\Delta_{\mathcal{L}}(\mathbf{x}_t) := \mathcal{L}_t(\mathbf{x}_t) - \tilde{\mathcal{L}}_t(\mathbf{x}_t)$, $\epsilon_t := \mathcal{L}_t(\mathbf{x}_t^*) - \mathcal{L}_t(\mathbf{x}_t)$,

where $\mathbf{x}_t^* = \arg \max_{\mathbf{x}} \mathcal{L}_t(\mathbf{x})$, $\mathbf{x}_t = \arg \max_{\mathbf{x}} \tilde{\mathcal{L}}_t(\mathbf{x})$,

Let $\mathbf{v}_t^* = v_{t-1} + \frac{1}{T} \mathbf{w}_t \odot \mathbf{x}_t^*$, $\mathbf{v}_t = v_{t-1} + \frac{1}{T} \mathbf{w}_t \odot \mathbf{x}_t$. Using Lemma

1 it is easy to see that $\Delta_{\mathcal{L}}(\mathbf{x}_t^*) = \Delta_f(v_t^*) \in O(\frac{1}{T^2})$, and $\Delta_{\mathcal{L}}(\mathbf{x}_t) = \Delta_f(v_t) \in O(\frac{1}{T^2})$. Since $\mathbf{x}_t^* = \arg \max_{\mathbf{x}} \mathcal{L}_t(\mathbf{x})$, $\mathbf{x}_t = \arg \max_{\mathbf{x}} \tilde{\mathcal{L}}_t(\mathbf{x})$, we have

$$\begin{aligned} \mathcal{L}_t(\mathbf{x}_t^*) &= \tilde{\mathcal{L}}_t(\mathbf{x}_t^*) - \Delta_{\mathcal{L}}(\mathbf{x}_t^*) \\ &\leq \tilde{\mathcal{L}}_t(\mathbf{x}_t) - \Delta_{\mathcal{L}}(\mathbf{x}_t^*) \\ &= \mathcal{L}_t(\mathbf{x}_t) + \Delta_{\mathcal{L}}(\mathbf{x}_t) - \Delta_{\mathcal{L}}(\mathbf{x}_t^*). \end{aligned}$$

Therefore, the approximation error $\epsilon_t = \mathcal{L}_t(\mathbf{x}_t^*) - \mathcal{L}_t(\mathbf{x}_t) \leq \Delta_{\mathcal{L}}(\mathbf{x}_t) - \Delta_{\mathcal{L}}(\mathbf{x}_t^*) \in O(\frac{1}{T^2})$, which completes the proof. $\square$

A.5 Proof of Regret Bound

A.5.1 Proof sketch of Theorem 5. Here we provide a sketch of proof including all major steps. The complete proof is shared at <https://github.com/Traiblazerss/DivFair>.

PROOF. $\text{Regret}(f) = W_{\text{OPT}} - W_{\text{online}} = \sum_{t=1}^T (W_{\text{opt}}^t - f(v_t))$, where $v_t = v_{t-1} + \frac{1}{T} \mathbf{w}_t \odot \mathbf{x}_t$ is calculated using $\mathbf{x}_t$ from Algorithm 1. With the same notations with Theorem 4, we can show according to weak duality,

$$W_{\text{opt}}^t - f(v_t) \leq \max_{\mathbf{e}_t \leq \boldsymbol{\theta}_t} \boldsymbol{\mu}_t^\top (\mathbf{e}_t - \frac{1}{T} \mathbf{x}_t) + \epsilon_t = \boldsymbol{\mu}_t^\top \mathbf{g}_t + \epsilon_t. \quad (11)$$

Theorem 4 tells us $\sum_{t=1}^T \epsilon_t \in O(\frac{1}{T})$. Next we show the bound for $\sum_{t=1}^T \boldsymbol{\mu}_t^\top \mathbf{g}_t$.

From mirror decent algorithm we have $\|\boldsymbol{\mu}_{t+1} - \boldsymbol{\mu}_t\| \leq \eta \|\mathbf{g}_t\| \leq \eta G$. Let $\mathbf{x} = \boldsymbol{\mu}_t - \eta \mathbf{g}_t$, $\mathbf{z} = \boldsymbol{\mu}_{t+1} = \Pi_{\mathcal{U}}(\mathbf{x})$. The first-order necessary optimality condition gives

$$(\boldsymbol{\mu} - \boldsymbol{\mu}_{t+1})^\top [(\boldsymbol{\mu}_t - \eta \mathbf{g}_t) - \boldsymbol{\mu}_{t+1}] \leq 0.$$

Rearranging and letting $\mathbf{a} := \boldsymbol{\mu}_t - \boldsymbol{\mu}_{t+1}$, $\mathbf{b} := \boldsymbol{\mu} - \boldsymbol{\mu}_{t+1}$ leads to

$$(\boldsymbol{\mu}_t - \boldsymbol{\mu})^\top \mathbf{g}_t \leq \mathbf{a}^\top \mathbf{g}_t - \frac{1}{\eta} \mathbf{a}^\top \mathbf{b}. \quad (12)$$

We now bound each inner product. By Young's inequality [59] and using $\mathbf{a}^\top \mathbf{b} = \frac{\|\mathbf{b}\|_2^2 - \|\mathbf{b} - \mathbf{a}\|_2^2 + \|\mathbf{a}\|_2^2}{2}$, hence,

$$-\frac{1}{\eta} \mathbf{a}^\top \mathbf{b} = -\frac{1}{2\eta} (\|\boldsymbol{\mu} - \boldsymbol{\mu}_t\|_2^2 - \|\boldsymbol{\mu} - \boldsymbol{\mu}_{t+1}\|_2^2 + \|\mathbf{a}\|_2^2). \quad (13)$$

Combining (13) and (12) yields the bound

$$(\boldsymbol{\mu}_t - \boldsymbol{\mu})^\top \mathbf{g}_t \leq \frac{\|\boldsymbol{\mu} - \boldsymbol{\mu}_t\|_2^2 - \|\boldsymbol{\mu} - \boldsymbol{\mu}_{t+1}\|_2^2}{2\eta} + \frac{\eta}{2} \|\mathbf{g}_t\|_2^2, \quad (14)$$

for any $\boldsymbol{\mu} \in \mathcal{U}$. Summing up (14) from $t = 1$ to T and take $\boldsymbol{\mu} = \mathbf{0}$ then

$$\sum_{t=1}^T \boldsymbol{\mu}_t^\top \mathbf{g}_t \leq \frac{\|\boldsymbol{\mu}_1\|_2^2}{2\eta} + \frac{\eta}{2} \sum_{t=1}^T \|\mathbf{g}_t\|_2^2 \leq \frac{D^2}{2\eta} + \frac{\eta G^2 T}{2}. \quad (15)$$

Combining (15) with (11), if we set learning rate $\eta = O(\frac{1}{\sqrt{T}})$, we obtain $O(\sqrt{T})$ bound on the regret. $\square$

A.5.2 Regret bound comparison.

COROLLARY 1. *DivFair has a strictly tighter regret bound than P-MMF, although they belong to the same order of magnitude.*

PROOF. We use superscript D and P to represent DivFair algorithm and P-MMF algorithm, respectively. Since both gradients $\|g_t^D\|$ and $\|g_t^P\| \in O(1)$, there exists a common constant $G \in \mathbb{R}^+$ such that $\|g_t\| \leq G$ for both algorithms.

For DivFair we obtain

$$\text{Regret}_D \leq \frac{D^2}{2} \sqrt{T} + \frac{G^2}{2} \sqrt{T} + O\left(\frac{1}{T}\right),$$

so the coefficient of $\sqrt{T}$ term is $C_D = \frac{D^2+G^2}{2}$.

For P-MMF we have

$$\text{Regret}_P \leq H\sqrt{T} + \frac{G^2}{c} \sqrt{T} + \frac{G^2 \sigma}{2c^2} \sqrt{T} + O(1),$$

hence $C_P = H + \frac{G^2}{c} \left(1 + \frac{\sigma}{2c}\right)$, where $c = (1 - \alpha) \sigma$. Therefore

$$C_P = H + \frac{G^2}{c} \left(1 + \frac{1}{2(1-\alpha)}\right) \geq H + \frac{G^2}{c} \left(1 + \frac{1}{2}\right) = H + \frac{3G^2}{2c}.$$

Moreover in practice $0 < c \leq 1$ (since $\alpha \leq 1$ and $\sigma \leq 1$), so $C_P \geq H + \frac{3G^2}{2} \geq \frac{3G^2}{2}$.

In DivFair, the gradient descent $\|\mu_{t+1} - \mu_t\| \leq \eta G$ gives $C_D \leq G^2$. Therefore, $C_D < C_P$ for all admissible (α, σ) , which gives a strictly tighter regret bound. $\square$

A.6 Comparing Variance of Dual Variables

In this section, we compare DivFair with a representative re-ranking model P-MMF, which targets provider fairness in online recommendation. We adapt P-MMF to individual-item level fairness and compare the variance of dual variables in two models. We prove that DivFair leads to more stable convergence.

Notations and assumptions. We use superscript D and P to represent DivFair and P-MMF, respectively. Based on widely adopted assumptions in large-scale RS [15, 27], let S_t be the size- K set recommended at round t . Let $\pi_{t,i}^D := \Pr(i \in S_t \mid S_{t-1}^D)$, $\pi_{t,i}^P := \Pr(i \in S_t \mid S_{t-1}^P)$. $x_{t,i}$ is a Bernoulli variable which equals 1 with probability $\pi_{t,i}$ and 0 otherwise. Following Lemma 2 in [13], we denote the maximum and minimum variance of $x_{t,i}$ by $v_{\max}$ and $v_{\min}$. In P-MMF algorithm, $\gamma_{\max}$ and α denote the maximum resource and momentum coefficient, respectively. $\text{Var}[\mu_{T,i}^D]$ and $\text{Var}[\mu_{T,i}^{P,\perp}]$ denote the variance of dual variables in DivFair and P-MMF, respectively.

THEOREM 6 (LOWER VARIANCE). *Let $A := \frac{\alpha}{2-\alpha} \left(1 - \frac{1}{I}\right) \frac{v_{\min}^P}{\gamma_{\max}}$, $B := \left(1 - \frac{1}{I}\right) \frac{v_{\min}^P}{\gamma_{\max}^2 \alpha}$, $C := v_{\max}^D$, and $T_* := \frac{B + \sqrt{B^2 + 4AC}}{2A}$. Then for every item i , when $T > T_*$, we have:*

$$\text{Var}[\mu_{T,i}^D] < \text{Var}[\mu_{T,i}^{P,\perp}]. \quad (16)$$

Remark. All quantities $K, I, M_f, \sigma, \gamma_{\max}, \alpha$ are observable before training; thus T_* is a closed-form numerical constant. Typically, based on Steam dataset, we have $I = 8000, K = 20, \alpha = 0.60, \gamma_{\max} = 1, \sigma = 0.30, v_{\max}^D = 0.25, v_{\min}^P = 0.18$, then $T_* \approx 9.2$ rounds before dual variable in DivFair gains smaller variance than P-MMF. We give a proof sketch of theorem 6 in Appendix A.6.1. The complete proof can be found at <https://github.com/Trailblazerss/DivFair>.

A.6.1 Proof sketch of Theorem 6. We first compute the variance of dual variables in P-MMF.

LEMMA 2 (P-MMF VARIANCE). *Under the P-MMF updates we have*

$$\text{Var}[\mu_{T,i}^{P,\perp}] = \eta^2 \kappa_T(\alpha) \Phi_i^P,$$

where

$$\kappa_T(\alpha) := \frac{\alpha}{2-\alpha} T - \mathcal{E}(\alpha, T), \quad 0 < \mathcal{E}(\alpha, T) < \alpha e^{-2\alpha T},$$

$$\Phi_i^P := \left(1 - \frac{1}{I}\right)^2 \frac{\bar{v}_i^P}{\gamma_i^2} + \frac{1}{I^2} \sum_{j \neq i} \frac{\bar{v}_j^P}{\gamma_j^2}, \quad \text{and} \quad \bar{v}_i^P := \frac{1}{T} \sum_{t=1}^T v_{t,i}^P.$$

The proof of Lemma 2 can be found at <https://github.com/Trailblazerss/DivFair>.

Smaller variance. Following the same notations with the above section, we have

$$\bar{v}_i^P \geq v_{\min}^P, \quad \frac{1}{T} \sum_{t=1}^T v_{t,i}^D \leq v_{\max}^D, \quad \gamma_i \geq \gamma_{\min}.$$

From DivFair, we can prove

$$\text{Var}[\mu_{T,i}^D] \leq \eta^2 \sum_{t=1}^T \text{Var}[g_{t,i}^D] = \eta^2 \frac{1}{T^2} \sum_{t=1}^T v_{t,i}^D \leq \eta^2 \frac{v_{\max}^D}{T}.$$

From the exact variance-gap formula $\Delta_{T,i} := \text{Var}[y_{T,i}^{P,\perp}] - \text{Var}[\mu_{T,i}^D]$ we obtain from Lemma 2 that

$$\Delta_{T,i} \geq \eta^2 \left[\left(\frac{\alpha}{2-\alpha} T - \frac{1}{\alpha} \right) \left(\left(1 - \frac{1}{I}\right)^2 \frac{v_{\min}^P}{\gamma_{\max}^2} + \frac{I-1}{I^2} \frac{v_{\min}^P}{\gamma_{\max}^2} \right) - \frac{v_{\max}^D}{T} \right].$$

Noting that $\left(1 - \frac{1}{I}\right)^2 + \frac{I-1}{I^2} = 1 - \frac{1}{I}$, Hence $\Delta_{T,i} \geq \eta^2 \left(AT - B - \frac{C}{T} \right)$, where $A := \frac{\alpha}{2-\alpha} \frac{v_{\min}^P}{\gamma_{\max}^2} \left(1 - \frac{1}{I}\right) > 0$, $B := \left(1 - \frac{1}{I}\right) \frac{v_{\min}^P}{\gamma_{\max}^2 \alpha}$, $C := v_{\max}^D$.

To ensure $\Delta_{T,i} > 0$, we require $AT - B - \frac{C}{T} > 0$, which holds exactly for $T > \frac{B + \sqrt{B^2 + 4AC}}{2A} =: T_*$. and the variance gap $\Delta_{T,i} > 0$ for all $T > T_*$. This completes the proof.

A.7 Faster Convergence of Dual Variables

In online optimization, sub-gradient $g_{t,i} > 0$ means item i is overexposed in this round and underexposed otherwise. Theorem 7 says DivFair drives this exposure gap to zero faster than P-MMF item i , which, together with Theorem 6, shows that stochastic noise in the update of dual variable is much smaller in DivFair.

THEOREM 7 (LOWER ACCUMULATIVE ERROR BOUND). $\forall \delta > 0$,

$$\Pr \left[\forall i, \left| \frac{1}{T} \sum_t g_{t,i}^D \right| \leq C_D \frac{\sqrt{\ln(I/\delta)}}{T^{3/2}} \right] \geq 1 - \delta,$$

$$\Pr \left[\forall i, \left| \frac{1}{T} \sum_t g_{t,i}^P \right| \leq C_P \frac{\sqrt{\ln(I/\delta)}}{T^{1/2}} \right] \geq 1 - \delta,$$

where C_D, C_P are explicit functions of $\alpha, v_{\max}^D, v_{\max}^P$.

Theorem 7 can be proved using Freedman's inequality [20]. The complete proof can be found at <https://github.com/Trailblazerss/DivFair>.

In practice, after only a few dozen rounds, DivFair algorithm is already nearly budget-balanced for items, whereas P-MMF algorithm still shows appreciable systematic over- or under-exposure.